DATE:

EX: 5 DEVELOP A POPULATION GROWTH MODEL

AIM

To develop a simple mathematical model for population growth using Python and use the model to predict future population.

ALGORITHM

1. Start the program.
2. Import NumPy and Matplotlib.
3. Enter the population data for the years 2020 to 2024.
4. Convert the years into time values starting from zero.
5. Plot the actual population data.
6. Fit a linear population growth model using np.polyfit().
7. Calculate the predicted population for the given years.
8. Predict the population for 2025, 2026 and 2027.
9. Plot the actual, model and forecast population values.
10. Display the results and stop the program.

DATE:

PROGRAM

```
import numpy as np
import matplotlib.pyplot as plt

year = np.array([2020, 2021, 2022, 2023, 2024])
population = np.array([10000, 10500, 11000, 11600, 12000])

t = np.array([0, 1, 2, 3, 4])

plt.figure(figsize=(10, 5))
plt.plot(year, population, marker='o', label='Actual')

b, a = np.polyfit(t, population, 1)
predicted = a + b * t

print("Starting Population =", a)
print("Growth per Year =", b)

print("\nYear Actual Predicted")
for i in range(len(year)):
    print(year[i], population[i], round(predicted[i], 2))

future_year = np.array([2025, 2026, 2027])
future_t = np.array([5, 6, 7])

future_population = a + b * future_t

print("\nFuture Population")
for i in range(len(future_year)):
    print(future_year[i], round(future_population[i], 2))

plt.plot(year, predicted, marker='x', linestyle='--', label='Model')
plt.plot(future_year, future_population, marker='s',
         linestyle=':', label='Forecast')

plt.title("Population Growth Model")
plt.xlabel("Year")
plt.ylabel("Population")
plt.legend()
plt.grid()
plt.show()
```

DATE:

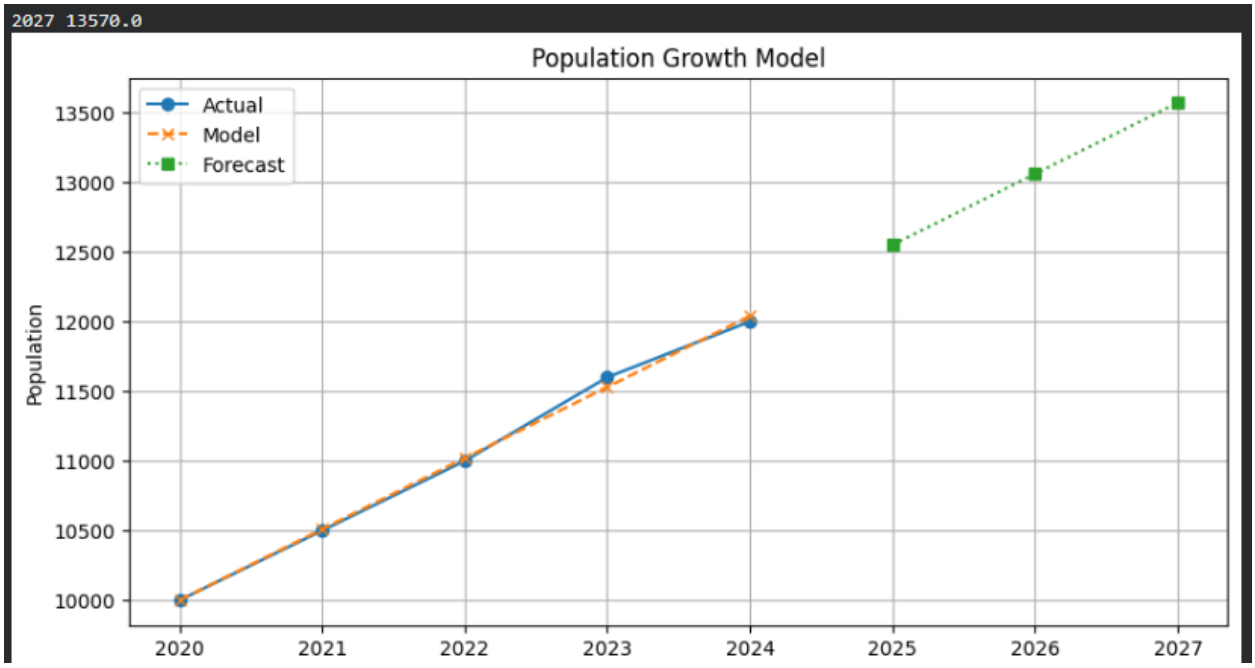
OUTPUT

```
Starting Population = 10000.000000000002
Growth per Year = 509.99999999999915

Year Actual Predicted
2020 10000 10000.0
2021 10500 10510.0
2022 11000 11020.0
2023 11600 11530.0
2024 12000 12040.0

Future Population
2025 12550.0
2027 13570.0
```

GRAPH



OBSERVATION

The population shows a steady increasing trend. The fitted linear model closely follows the actual values and predicts continued growth for 2025, 2026 and 2027.

RESULT

Thus, a simple population growth model was developed using Python. The model was used to estimate the population for future years and visualize the population growth trend.